Energy Consumption of a Building by using Long Short-Term Memory Network: A Forecasting Study

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***Abstract*—Buildings have a dominant presence in energy consumption for the transition to clean energy. During 2017, construction and operation of buildings worldwide represented more than a third (36%) of final energy used and 40% of the emissions of carbon dioxide. Hence, there is great interest in reducing energy use in this sector, and energy efficiency in buildings to enhance energy performances is a suitable way. In this paper, black-box approaches based on artificial neural networks to predict the electric load of a selected educational building are proposed. The potential and robustness of long short-term memory (LSTM) applied to a dataset with a limited number of days of observations are analyzed. The results in our scenario showed that the LSTM surpasses in accuracy to other techniques such as feed-forward neural networks.**

Keywords—energy efficiency building, electric load time series, long short-term memory, feed-forward neural networks.

I. Introduction

Energy consumption in buildings represents a dominant role in the transition to clean energy. International Energy Agency report shows increasing trends in energy consumption and energy-related carbon emissions used in the building construction sector. During 2017, buildings worldwide' construction and operation represented more than a third (36%) of the final energy use. The emissions of carbon dioxide (a greenhouse gas) related to the energy used by existing buildings, manufacture of materials, and products for the construction of new buildings accounted for about 40% of the global world emissions in the same year [1].

According to the National Energy Balance 2018, the national consumption in residential, commercial, services, and public administration buildings was approximately 21% of the total consumption in the Ecuadorian case. The primary source of energy used by buildings was electricity, representing 57.6%, i.e., 29.7% residential group and 27.9% for the commercial, services, and public administration group. The demand for total energy in 2018 was 3.8% higher than in 2017. The same behavior has been observed in the historical years, i.e., the demand for energy in Ecuador is increasing annually [2].

Considering the previously mentioned scenario, the Ecuadorian government promotes energy efficiency through a law approved in March 2019. Among the regulated sectors by this law, energy efficiency in buildings (EEB) is considered as part of the regulation [3]. The EEB is a field of research that has been greatly boosted by interdisciplinary research groups. In Ecuador, some studies have been performed in a selected university administration building [4].

EEB mainly seeks to optimize designs and to enhance energy performances. There is a research field related to building modeling and energy performance prediction. From a research point of view, two approaches are commonly considered: The physical field (white box), oriented to solve equations that simulate the thermal behavior of the building, and the mathematical field (black box), which consists in the implementation of a prediction model based on machine learning techniques [5].

EEB has received massive attention in most countries during the last years. For new buildings, preliminary designs based on zero energy building (ZEB) are required. On the other hand, existing buildings are being modernized and reduce their energy consumption through a sustainable model of building energy efficiency retrofit (BEER) [6]. Possible solutions for existing buildings include efficient energy management and smart refurbishments. Therefore, an accurate energy prediction is required for optimal decision making [7].

In addition, a bibliographic study about data analysis techniques applied in EEB [8] shows that in the last ten years, there has been exponential growth in publications of this line of research. Data analysis techniques mostly used are the following: Regression analysis, descriptive statistics, multivariate analysis, computational intelligence, stochastic processes, inferential statistics, and design of experiments.

In this paper, black-box approaches based on artificial neural network (ANN) to investigate the application of Long Short-Term Memory (LSTM) for energy time series forecasting is applied. The performance of LSTM when there is not a large amount of energy dataset of an educational building is analyzed. For this purpose, contrasts are made with other techniques such as a feed-forward neural network (FNN).

The rest of the paper is organized as follows. Section II summarizes the related works. In section III, a brief explanation of ANN techniques is presented. Section IV shows the experiment evaluations and discussion of the results. Finally, Section V includes some concluding remarks and directions for future work.

II. Related Works

ANN rank among the most popular data-driven approaches applied to forecast the building energy consumption [9]. A comparative analysis of ANN architectures for building energy consumption forecasting is shown in [10]. It also highlights the importance of accurately forecasting the electrical load of buildings. Experimental results of the cited study indicate that ANN with

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hyperparameters modified outperforms other forecasting models.

Some applications of ANN in different types of buildings, according to their use, are found in the literature. A study in [11] considered office buildings in different climates, and its target is to predict the energy consumption of heating, ventilating, and air conditioning (HVAC) systems. Another case of office buildings in [12] is presented, in which the total energy consumption of the building is disaggregated in HVAC systems, lights, and electrical sockets. The study presented in [13] considers an analysis of the building envelope's thermal behavior and other characteristics to predict the energy performance of residential buildings. An energy consumption forecasting tool based on ANN and applied in a public building is detailed in [14].

There are also related works involving LSTM. The study presented in [15] uses LSTM-based deep recurrent neural network models applied to commercial and residential buildings for predicting electricity consumption. A hybrid model of energy consumption forecasting using LSTM, named eDemand, is used in [16]. Due to its capacity to learn long-term dependencies, LSTM can be used to predict power consumption and detect anomalies such as theft of electricity [17], [18].

In this paper, an extension of previous work is detailed by investigating the potential of LSTM to forecast building energy consumption in an educational building. It is also compared to multilayer perceptron FNN.

III. Artificial Neural Network Techniques

ANN is a non-linear computational model inspired by the human brain. In general, the structure of an ANN is made up of three sequential layers: the input layer, the hidden layer, and the output layer. An ANN can have multiple levels of hidden layers. Each layer has several interconnected neurons, and each neuron has an activation function. There are commonly three parameters that define an ANN: i) the interconnection pattern between the neurons of the different layers; ii) the learning process of updating the weights of the interconnections, and iii) the activation function that converts a neuron's weighted input to its output activation [19].

The input layer consists of the regressor variable(s) named features used to estimate the target variable (prediction). The connection between the layers is weighted through the weights found during the neural network's training phase. Many different training methods such as backpropagation, genetic algorithm, Levenberg-Marquardt algorithm are available in the literature. The training goal is to minimize the error between the ANN output and target output, given a set of known inputs and target values [9].

A recurrent neural network (RNN) and Feed-forward neural network (FNN) are considered in this work. Both mentioned networks are examples of ANN.

A. Feed-forward neural network

FNN is a type of ANN in which the computations proceed only in the forward direction. FNN is often called multilayer perceptron (MLP) or backpropagation neural network (BPNN). It is essential to notice that the convolutional neural network (CNN) and deep neural network (DNN) are classified as FNN. Fig.1 depicts a typical FNN topology.

During the last years, FNN has achieved updated performance in various supervised or unsupervised machine

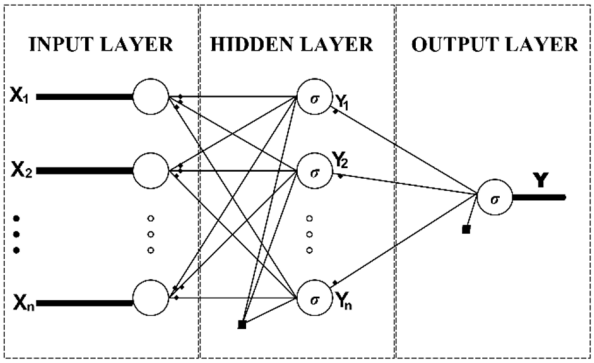


Fig. 1. A typical FNN topology

learning applications. Its success highly depends on the independence assumption among the training and test data [20].

In this study, the performance of RNN-LSTM with a standard one hidden layer FNN-MLP is analyzed.

B. Recurrent neural network

A time series is a set of samples taken at regular time intervals. Time series analysis comprises a vast collection of techniques for analyzing historical temporal data to extract meaningful features or characteristics of the data. Time series have two essential characteristics: i) their observations depend on time and, ii) they usually have the characteristic of seasonality and do not fulfill stationarity [21], [22].

According to the mentioned features, there would be a particular limitation of using FNN in time series problems. In addition, FNN can only be applied to problems whose inputs and targets are vectors of fixed dimensionality. It is a significant limitation since many important problems are better expressed with sequences (such as time series) whose lengths are not known a-priori [23].

Hence, RNN is the alternative to deal with sequential data or time series. Unlike FNN that only allows signals to travel forward from the input to the output, RNNs allow signals to travel both forward and backward. Therefore, they are more powerful and robust since they have a fully connected structure of neurons with internal memory and cycles that loop back the information of the previous time step into the network, i.e., information sharing between the time steps [16].

However, the backpropagation algorithm is usually used to train a neural network and propagate a feedback signal from the output loss down to earlier layers. If this feedback signal has to be propagated through a deep stack of layers, the signal may become tenuous or even be lost entirely, rendering the network untrainable. This issue is known as the vanishing gradients problem (VGP). This problem occurs over very long sequences (long-term dependencies) in FNN and RNN [24]. Despite its benefits, RNN suffers from VGP, and therefore it has difficulty in learning long-term dependencies.

C. Long-Short-Term-Memory model

LSTMs can learn short- and long-term temporal dependencies in time series data. Hochreiter and Schmidhuber designed this new architecture for RNN based on the concept of "memory blocks or gate units" in each hidden layer. They

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had as main motivation to solve VGP [25]. Figure 2 shows a schematic diagram of an LSTM unit.

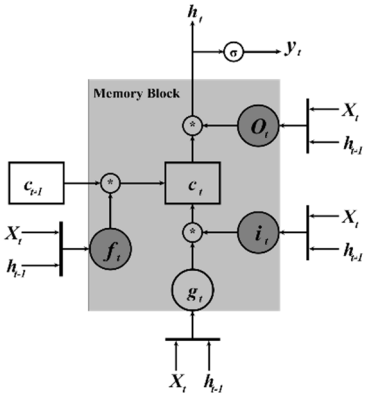


Fig. 2. Schematic diagram of a LSTM unit

Gate unit is a linear self-loop memory block that allows gradients to flow through long sequences. The memory block that contains self-connecting memory cells c_t can memorize the temporal state, and three gates within the unit can help to control the flow of information inside and outside of the memory block: i) forgetting gate f_t, which conditionally decides the information to discard from the block; ii) input gate i_t, which conditionally decides which values from the input update the memory state; and, iii) output gate o_t, which conditionally decides the output based on input and the memory of the block. Therefore, each unit is like a mini-state machine where the units' gates have weights that are learned during the training process [18], [26].

Due to its inherent capability in modeling temporal aspects of data, LSTM has been widely employed in machine translation, language modeling, handwriting recognition, image captioning, speech recognition, process prediction, and medical diagnosis [26].

In this work, LSTM for electric load time series forecasting of an educational building is employed. Given an input time series $x = \{x_1, x_2, \cdots, x_T\}$, the LSTM maps input time series to two output time sequences $h = \{h_1, h_2, \cdots, h_T\}$ and $y = \{y_1, y_2, \cdots, y_T\}$ iteratively by updating the states of memory cells with the following procedure [20], [26]:

Each gate is a sigmoid unit that change every element in [0, 1], i.e., they use a logistic function defined in eq. (1):

$$\sigma(x) = \frac{1}{1 + e^{-x}} \tag{1}$$

Input gate i_t controls the input of new information that is going to be stored in the new cell state, which derives the following:

$$i_t = \sigma(w_{xi}x_t + w_{hi}h_{t-1} + b_i) \tag{2}$$

Forgetting gate f_t decides what information to throw away from cell state, where:

$$f_t = \sigma\left(w_{xf}x_t + w_{hf}h_{t-1} + b_f\right) \tag{3}$$

Output gate o_t controls and filters the output information flowing out of the cell, where:

$$o_t = \sigma(w_{xo}x_t + w_{ho}h_{t-1} + b_o) \tag{4}$$

At each time t, the input features are computed by input x_t and the previously hidden state h_{t-1} by using a *tanh* function, as follows:

$$g_t = tanh(w_{xc}x_t + w_{hc}h_{t-1} + b_c) \tag{5}$$

The memory cell is updated by moderated input features and the partial forgetting of the last memory cell, which yields:

$$c_t = f_t * c_{t-1} + i_t * g_t \tag{6}$$

Then, the hidden output state h_t is calculated by output gate o_t and memory cell c_t, as follows:

$$h_t = o_t * \tanh(c_t) \tag{7}$$

Finally, the output of LSTM y_t is computed by using the following expression:

$$y_t = \sigma\left(w_{hy}h_t + b_y\right) \tag{8}$$

In Eqs. (2)-(8), the matrices w_{xi}, w_{xf}, w_{xo}, and w_{xc} are the appropriate input weighting matrices, while w_{hi}, w_{hf}, w_{ho}, and w_{hc} are the recurrent weighting matrices; and w_{hy} represents the hidden output weight matrix. The vectors b_i, b_f, b_o, b_c, and b_y are the corresponding bias vectors.

IV. Experiment Evaluations and Discussion

In this section, the experiment's setup and results achieved with FNN-MLP and RNN-LSTM models are presented and discussed.

A. Experiment setup

Data: An energy dataset with observations of some variables collected from an educational building of the University of Guayaquil is used in the current study. They were collected with high precision equipment: FLUKE 435 II during two days of February 2020, with a frequency of 1/2 - minute, i.e., the dataset has 5760 measurements.

ANN has proven to be applicable in time series analysis. Its forecast accuracy depends, among other factors, on the quality and quantity of data that will be used to train and validate the model. Based on bibliographic references, datasets with hundreds of thousands of observations for training and validation of neural networks are commonly used.

In this paper, the potential of LSTM considering the dataset with two days of measurements is analyzed. The total electric load of the building or total power consumption is considered during the analysis. Figure 3 depicts the total

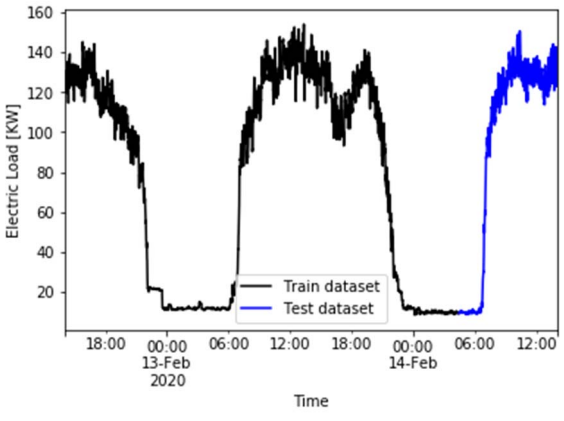


Fig. 3. Electric load curve of the analyzed building

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electric load curve behavior. In the university building analyzed in this study, the first hour of classes starts at 7:00 am and the last hour of class finishes at 10:00 pm. Its peak electric load is about 152 kW. The fifteen hours of daily operation represents an approximated electrical consumption of 1835.97 kWh.

Problem Framing: Our research question in this work is: Given a recent power consumption, what is the expected power consumption for one step ahead? That is, given historical observations (*t*-1, *t*-2, … *t*-n), forecast the value *t*.

The data is resampled to 5-minutes observations of power consumption. The 5-minutes resolution data were obtained by averaging the 1/2-minute resolution data (original dataset).

On the other hand, the dataset is divided into train and test data, considering the proportion of 80% and 20%, respectively.

Normalization is applied to the data sets, and they are transformed into values between -1 and 1.

Model Evaluation: As it is known, k-fold cross-validation does not work for time series data because they ignore its temporal relation. Therefore, a rolling-forecast scenario will be used, also called walk-forward model (WFM) validation. Each time step of the test dataset will be walked one at a time [27].

Performance evaluation: In literature reviews in which model adjustments with energy observations are handled, the Root Mean Squared Error (RMSE) metric is the most used evaluation parameter. Therefore, the RMSE is considered the main metric to select the best architecture with the lowest forecast error. Other supporting metric is the Mean Absolute Error (MAE) [11], [16], [26], [28] .

$$RMSE = \sqrt{\frac{1}{N_{test}} \sum_{n=1}^{N_{test}} (\tilde{y}_n - y_n)^2} \tag{9}$$

$$MAE = \frac{1}{N_{test}} \sum_{n=1}^{N_{test}} |\tilde{y}_n - y_n| \tag{10}$$

where N_{test} is the number of test set samples, y_n refers to the real value of the *nth* forecasting point, and $\tilde{y}_n$ is the corresponding predicted value.

Experiment models: MLP and LSTM architectures for one-step time series forecasting with one hidden layer have been configured. Table 1 shows the variation of hyperparameters used in each of the executed models.

TABLE 1. HYPERPARAMETERS CONSIDERED FOR EXPERIMENT MODEL

Hyperparameter	Values
n_input	1, 3, 4, 5, 6,7, 8, 9, 12, 24, 36
n_nodes	1, 2, 4, 6, 7, 8, 9, 10, 12, 16, 24, 28, 32, 36
n_epochs	50, 75, 100, 150, 200, 400, 500, 1000, 1500, 2000
n_batch	1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 20, 25
act_hid	tanh, relu, linear
act_out	tanh, linear

Where n_input is the number of prior inputs to use as input for the model, n_nodes is the number of nodes/units used in the hidden layer. n_epochs represent the number of training epochs, n_batch is the number of samples to include in each mini-batch. The activation function to be used in the hidden layer is given by act_hid, and act_out is the activation function used in the output layer.

Due to the stochastic nature of neural networks, each model is trained and validated ten times, then the mean of the RMSE errors is determined.

All experiment models were implemented in Python, using the TensorFlow library.

B. Experiment results with FNN-MLP

After carried out the experiments and fine-tuning hyperparameters using the train dataset and WFM validation in test dataset, the best topology neural network achieved is as follows: n_input = 1, n_nodes = 1 neuron, n_epochs = 100, n_batch = 6, act_hid = linear and act_out = linear.

Table 2 shows the top 10 FNN-MLP models based on the RMSE metric.

TABLE 2. TOP 10 FNN-MLP MODELS

Hyperparameters [n_input, n_nodes, n_epochs, n_batch, act_hid, act_out]	RMSE [kW]
[1, 1, 100, 6, 'linear', 'linear']	8.0322
[1, 2, 100, 6, 'linear', 'linear']	8.0712
[1, 2, 75, 6, 'linear', 'linear']	8.0875
[1, 1, 75, 6, 'linear', 'linear']	8.1044
[1, 4, 75, 6, 'linear', 'linear']	8.1773
[1, 4, 100, 6, 'linear', 'linear']	8.1951
[1, 4, 100, 6, 'relu', 'linear']	8.2132
[1, 4, 75, 6, 'tanh', 'linear']	8.2890
[1, 4, 100, 6, 'tanh', 'linear']	8.3148
[1, 12, 100, 6, 'linear', 'linear']	8.3317

Those with the least error in the validated models are the hidden layer with linear, tanh, and relu activation functions. There are seven models out of 10 with a linear activation function.

In the top 10 FNN-MLP models, RMSE errors range between 8.0322 and 8.3317, i.e., a range of 0.2995 kW. In addition, the following characteristics are observed:

The n_input values are constants, i.e., The number of nodes is 1, 2, 4, and 12. The epochs' values are 75 and 100. Finally, the common hyperparameters are n_batch = 6 and act_out = linear.

C. Experiment results with RNN-LSTM

As in the FNN-MLP, experiments and fine-tuning hyperparameters are carried out. It was found that the best architecture of RNN-LSTM is achieved under the following conditions: n_input = 8, n_nodes = 6 neurons (LSTM units), n_epochs = 300, n_batch = 1, act_hid = tanh and act_out = linear.

Table 3 shows the top 10 RNN-LSTM models based on the RMSE metric.

Those with the lowest error in the validated models are the hidden layer with ‘tanh’ activation function.

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In the top 10 RNN-LSTM models, RMSE errors range between 4.7312 and 5.0972, i.e., a range of 0.366 kW. In addition, the following characteristics are observed:

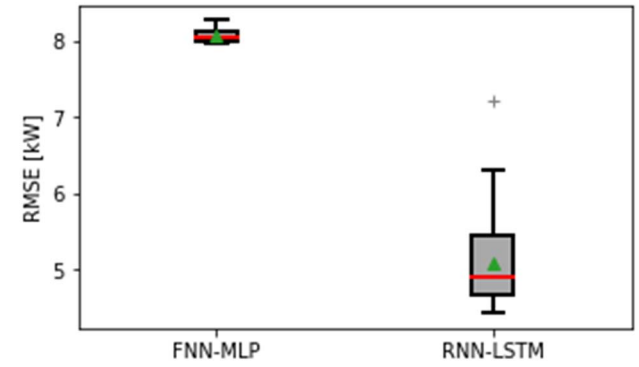
TABLE 3. TOP 10 RNN-LSTM MODELS

Hyperparameters [n_input, n_nodes, n_epochs, n_batch, act_hid, act_out]	RMSE [kW]
[8, 6, 300, 1, 'tanh', 'linear']	4.7312
[8, 7, 100, 1, 'tanh', 'linear']	4.8089
[8, 9, 200, 1, 'tanh', 'linear']	4.8259
[6, 8, 200, 1, 'tanh', 'linear']	4.8964
[8, 9, 100, 1, 'tanh', 'linear']	4.8994
[8, 7, 200, 1, 'tanh', 'linear']	4.9292
[8, 10, 100, 1, 'tanh', 'linear']	4.9330
[8, 6, 200, 1, 'tanh', 'linear']	4.9477
[7, 9, 100, 1, 'tanh', 'linear']	5.0905
[9, 8, 200, 1, 'tanh', 'linear']	5.0972

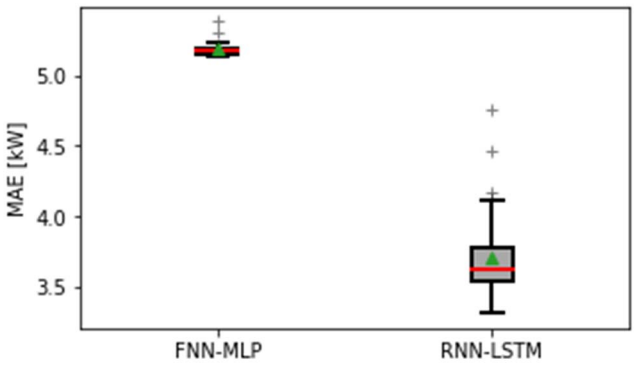
The n_input values vary between 6 and 9. There are seven models out of 10 with n_input = 8. The number of nodes varies between 6 and 10. The epochs' values are 100, 200, and 300. Finally, the common hyperparameters are n_batch = 1, act_hid = tanh, and act_out = linear.

For our case study, LSTMs have a good precision to predict the electrical load of the building. Comparing Tables 2 and 3, good learning behavior is evident. This is due to the long-term characteristics already mentioned in Section III.

As mentioned above in the experiment setup subsection, the results' randomness is manifested due to the stochastic nature of the neural networks. Figure 4 shows the box diagram of the RMSE and MAE errors of total power consumption of the building. The mentioned errors are obtained per each best architecture from 30 repetitions. It is visibly noted that LSTM outperforms FNN.



a)



b)

Fig. 4. Box plot errors of the experiments carried out with FNN and LSTM. a) RMSE and b) MAE. Red line represents the median while the green triangle corresponds to the mean.

RMSE: On average, the LSTM error is 5.085 kW, with a deviation of 0.625 kW. Moreover, 75% of the experiments (3 of 4 repetitions) fall below the 5.436 kW of electric load. In contrast, MLP error, on average, is 8.079 kW, with a deviation of 0.084 kW. Notice that 75% of the experiments fall below the 8.128 kW of electric load. LSTM error is 1.59 times lower than MLP.

MAE: On average, the LSTM error is 3.714 kW, with a deviation of 0.326 kW. Moreover, 75% of the repetitions fall below the 3.774 kW of electric load. In contrast, MLP error, on average, is 5.184 kW, with a deviation of 0.051 kW. In addition, 75% of the experiments fall below the 5.195 kW of electric load. LSTM error is 1.40 times lower than MLP.

V. CONCLUSIONS

In this study, an efficient LSTM is designed to forecast the electric load of a building. Experiments using a dataset of the electric load of an educational building were carried out. Two techniques of artificial neural networks were considered, and models were configured, trained, and validated. RMSE and MAE were used to select the best model for each type of architecture. Despite the limited number of observations in our case study, recurrent neural networks LSTM achieved better error metrics than Feed-Forward neural network techniques.

In further studies, it is expected to use LSTM in datasets with a more significant number of observations (at least one annual cycle) of the building referred. Moreover, forecasting problems such as indoor-outdoor temperatures, solar power forecasting, or complex demand forecasting using predictive models for energy management and control in buildings will be considered.

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